

Paper Practice 1.6 • WORKED KEY**Descriptive Modeling and Accuracy**

WORKED KEY (odd-numbered items) – Show all of your work in the space provided. When you use a metric, name it. Explanations should be one clear sentence. Circle your final answers.

- 1.** The school store's hoodie sales for five months were 42, 55, 38, 61, and 49. (Descriptive Modeling)

a. Find the mean number of hoodies sold per month.

Answer: $245 \div 5 = 49$ hoodies

b. Name one thing the mean HIDES about these sales.

Answer: The spread – sales swing from 38 to 61, so a typical month is not reliably 49.

c. Would a table or a bar graph be more useful for spotting a trend? Explain.

Answer: A bar graph – it shows the month-to-month rise and fall at a glance, while a table lists values without showing shape.

- 2.** Two players' season statistics are shown. (Compare Metrics)

Player	Points	Games
Alvarez	1,848	22
Brooks	1,530	17

- a.** Who leads in total points?
b. Who leads in points per game?
c. Which metric would you use to argue for each player? What does this show?

- 3.** A car travels 260 miles on 8 gallons of gas. Find the miles per gallon, and explain what that metric tells a buyer. (Metrics)

Answer: $260 \div 8 = 32.5$ miles per gallon – it lets a buyer compare fuel efficiency across cars regardless of tank size.

- 4.** A digital scale reads 9.985 grams for a mass whose true value is 10 grams. Find the accuracy of the reading. (Accuracy)

- 5.** A bag of flour labeled 5 pounds weighs 4.92 pounds. Find the accuracy, then express it as a percent of the labeled weight. (Accuracy)

Answer: $|4.92 - 5| = 0.08$ pound; $0.08 \div 5 = 0.016 = 1.6\%$ off the label

- 6.** Explain why accuracy is written with absolute value bars rather than plain subtraction. (Explain)

7. A principal wants to name the “most improved” class. Class A rose from 62 to 74; Class B rose from 80 to 88. (Choose a Metric)

a. Find each class's raw point gain. Who improved more?

Answer: A: $74 - 62 = 12$; B: $88 - 80 = 8$. Class A improved more.

b. Find each class's percent gain. Who improved more?

Answer: A: $12/62 \approx 19.4\%$; B: $8/80 = 10\%$. Class A again.

c. Both metrics agree here. Why does that make the conclusion stronger?

Answer: When two independent metrics point the same direction, the result does not depend on which metric you happened to pick.

8. Error Analysis – Gus says a thermometer reading of 101.2°F when the true temperature is 101.0°F has an accuracy of -0.2°F . Identify his mistake. (Error Analysis)

9. A team scored 1848 points in 22 games. Find the points per game, then explain why a coach might prefer this metric over total points. (Metrics)

Answer: $1848 \div 22 = 84$ points per game. It allows fair comparison between teams that have played different numbers of games.

10. Describe a real situation where the mean would be a POOR model, and name a better metric. (Reason)